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Department of Mining Engineering

Mid-Semester Examination (Winter Session) 2023-24 for B.Tech. Students (45 nos.)

Subject: Advanced Underground Mining Methods (MND 411) Venue: NLHC LH 08

Time: 120 Minutes (08.00 AM-10.00 AM) Date: 02.03.2024 Maximum Marks: 64

Answer questions of all the sections

Question No.	Section A	Marks
	Answer any four questions [4×10=40 Marks]	
1.	Discuss the operational challenges associated with thick seam mining. Enumerate the typical ground control requirements for mining thick seams.	[10]
2.	What are the different factors that govern the choice of face length in Longwall workings? Why it is critical to operate with optimal face length for a given mining condition?	[10]
3.	a. What are the different zones that form above the coal seam with the progressive mining in a typical longwall working? How do the dynamics of caving height alter with the variation of bulking factor and mining height? b. Discuss the supporting cycle and loading characteristics of the powered roof support deployed in the longwall operation.	[5+5]
4.	With neat sketches, describe the abutment stress distribution about the longwall panel.	[10]
5.	a. What are the typical loading scenarios in the longwall workings? b. Discuss the mechanism of face damage and various forms of damage parameters.	[3+7]

Section B

Answer all the questions [2×12=24 Marks]

7. a. Estimate the different roof layers above the coal seams for a longwall panel of Mine 'A' by applying the Roof Separation Index (RSI) approach (Refer to Table 1), where there is no influence of groundwater on the working. (Refer to Roof Separation Index (RSI) Table, Table 2, provided in the question paper for assignment of different ratings associated with RSI parameters) [5+7]
- b. Applying the concepts of Beam and Cantilever models, estimate the spans of the first local fall, subsequent local fall, main fall and periodic caving pertained to the working of Mine 'A'.

3015
120
82

8.

a. A longwall face of Mine 'B' is planned to operate under the cover depth of 150m with a face length and face height of 180m and 3.5m, respectively. The Physico-Mechanical properties of overlying roof rocks related to the working are given in Table 3. Estimate the capacity of powered roof support that needs to be deployed in the given condition considering the canopy length of 4m, canopy width of 1.75m and web depth of 0.8m for the working. Below are some useful relations for solving this question:

[7+5]

$$C_m = \left(\frac{1}{P_b} + 9.6h_s \right) + \left(\frac{K_1}{K_1 + 1.5} - 23 \right)$$

$$\text{Support Capacity (T)} = P_r \times W_c \times (L_c + W_d + 0.3)$$

b. Keeping the other parameters constant, If the mine management planned to operate the longwall face with a 4.5m face, then what would be the support capacity for altering the face height from 3.5m to 4.5m?

Table 2: Roof Separation Index (R.S.I) Table

Sl. No.	Parameter	Unit/ Rating	Rating distribution				
			<5	5-25	25-50	>50	
1	Uni-axial compressive strength	MPa Rating	<5 1	5-25 2	25-50 4	>50 7	
2	RQD	% Rating	<25 3	25-50 8	50-75 13	75-90 17	>90 20
3	Average core length	cm Rating	<5 5	5-30 10	30-100 20	100-300 25	
4	Ground water condition/wetness of strata	Condition Rating	Dry 10		Watery formation 0		
5	Bed thickness	m Rating	<1.5 -28	1.5-2.5 -16	2.5-3.5 -12	3.5-4.5 -5	>=4.5 0
6	Rock formation of bed	Rock type Rating	Sandstone 0		Others (shale, coal, clay, shaly coal) -25		